



SHADOW AI DATA GUARD

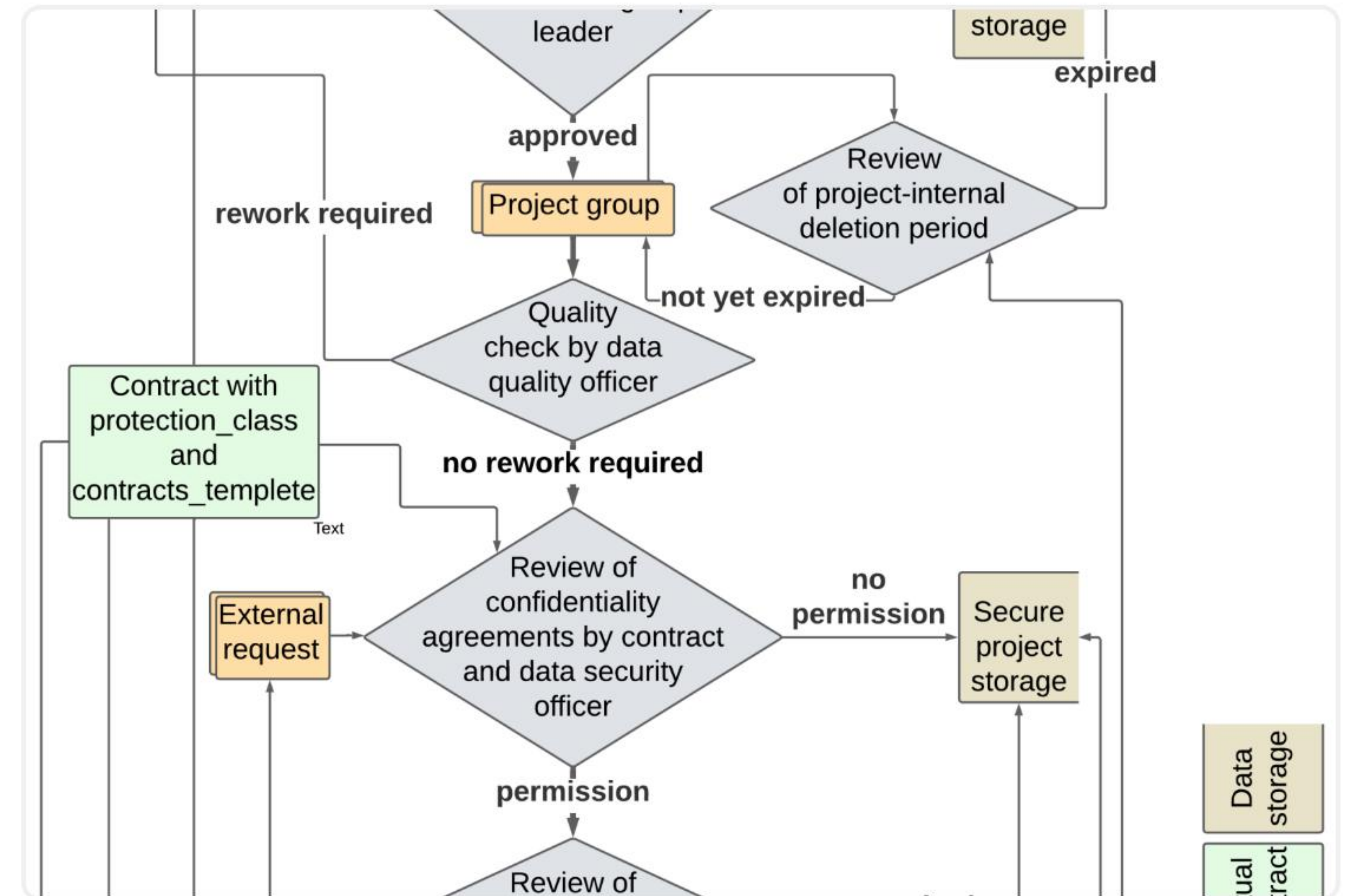
Experiential Learning Phase-I

Project Sector: AI in Cyber Security

Team Members: [Four Student Names]

 RVCE Campus

EL – OUTLINE OF TENTATIVE STEPS



STRATEGIC ROADMAP

01. Intro

Overview & Significance

02. Review

Literature Gap Analysis

03. Problem

Semantic Leakage Risk

04. Methodology

Design & Architecture

05. Outcome

Results & Timeline

06. EL Results

First Year Goals

UNDERSTANDING SHADOW AI

Overview

Shadow AI involves unsanctioned use of public GenAI (ChatGPT/Claude) by employees, leading to exfiltration of corporate secrets.

Importance

By 2026, 38% of employees have shared sensitive data with AI tools without permission. Justification lies in mitigating "Zero-Day" semantic leaks.



KEY STUDIES & FINDINGS

Publication	Contribution
IEEE (2025)	Evaluated BERT models for PII redaction accuracy.
ArXiv (2026)	Analyzed Shadow AI trends in enterprise security.
MDPI (2026)	Developed context-aware LLM guardrails.
Cisco (2026)	Identified semantic exfiltration as a top 2026 threat.



STATEMENT OF THE PROBLEM

The Core Issue

Standard firewalls lack **semantic awareness**. They cannot stop a user from copy-pasting proprietary code or PII into an LLM prompt.

Consequences

Lack of monitoring results in non-compliance with DPDP/GDPR and massive loss of corporate competitive advantage.



PRIMARY & SPECIFIC GOALS



Main Objective

Build a real-time semantic gateway for PII redaction.



NLP Focus

Optimize BERT-NER for 98% recall accuracy.



Low Latency

Ensure processing takes less than 500ms.

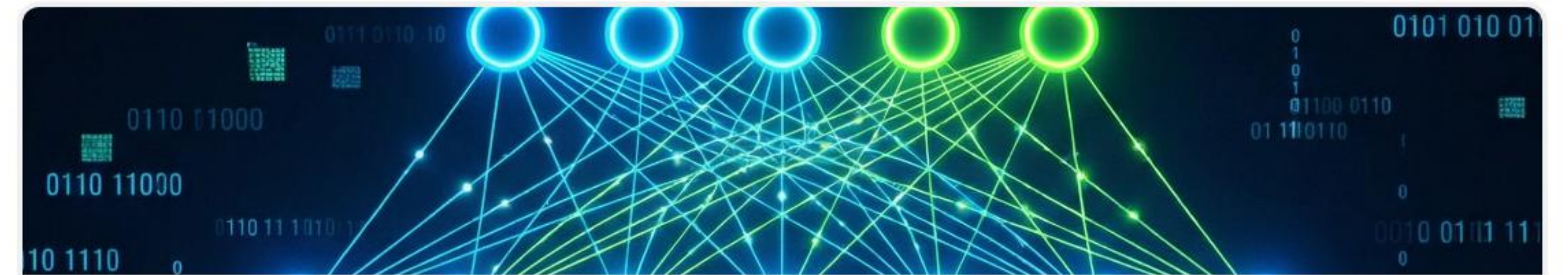
FRAMEWORK & TOOLS

Approach

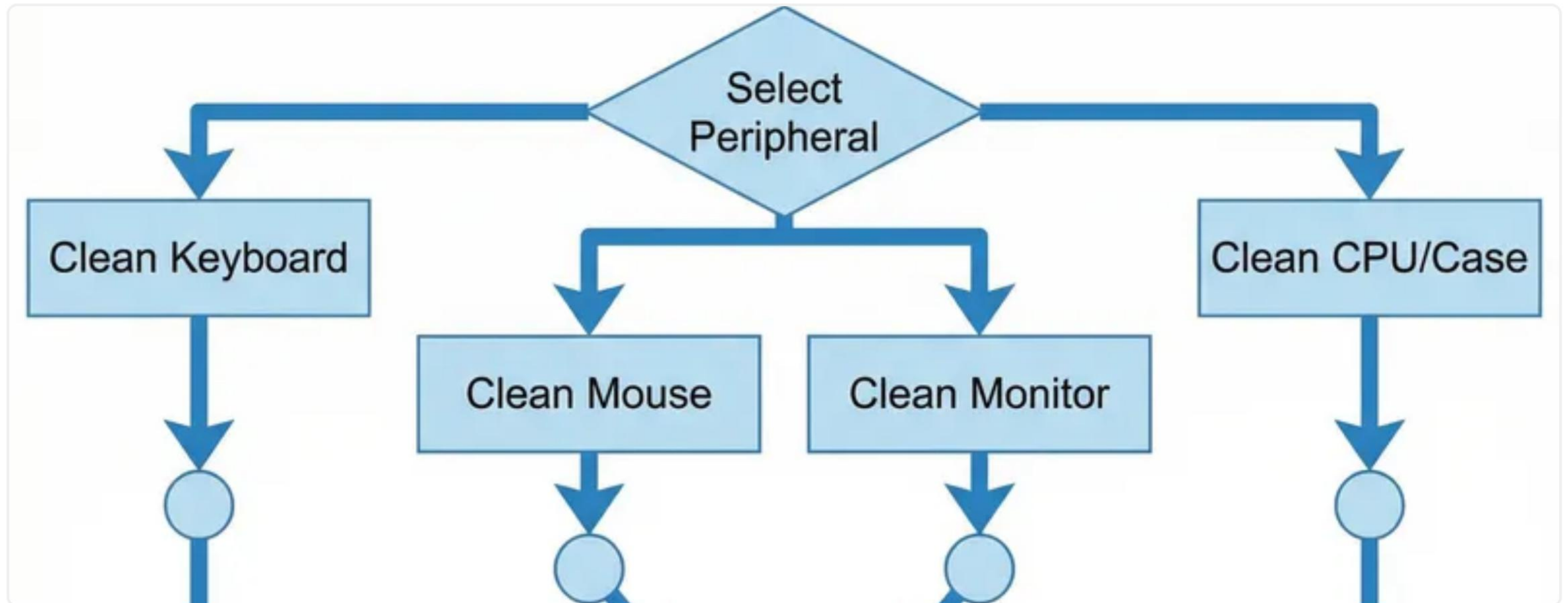
Quantitative analysis using Named Entity Recognition (NER).
We fine-tune BERT (Bidirectional Encoder Representations from Transformers).

Resources Used

Python 3.12, SpaCy, HuggingFace Transformers, and Streamlit for interface development.



SYSTEM RESEARCH PROCESS



Flow: Input Hook -> BERT-NER Detection -> Redaction Layer -> Sanitized API Call

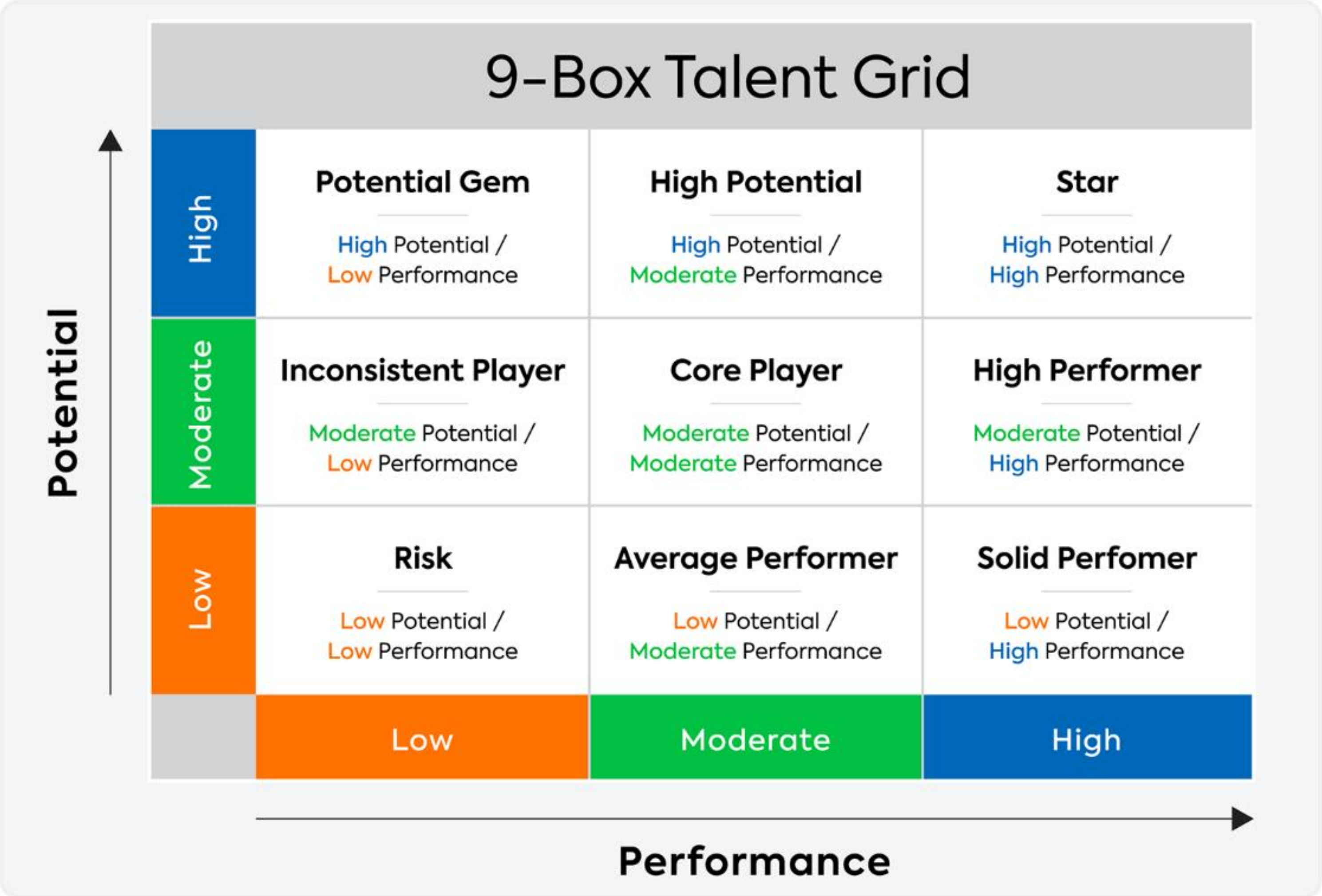
EXPECTED RESULTS & IMPACT

99%

Reduction in accidental data leaks via public AI channels.

Future Directions

Expanding model to multi-modal data (images/PDF uploads).



OVERVIEW OF PHASES

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- Initiation (Weeks 1-2): Team formation & Feasibility study.
 - Planning (Weeks 3-4): Detailed budget & task allocation.
 - Execution (Weeks 5-10): Model coding & integration meetings.
 - Closure (Weeks 11-13): Deliverables & Lessons learned.

FINAL THOUGHTS

Our project addresses the critical gap in Shadow AI governance. By using autonomous redaction, we enable safe innovation without compromising data sovereignty.

Roadmap

- Phase II: Corporate beta testing.
- Phase III: Scaling to distributed cloud nodes.

ACADEMIC CITATIONS

1. Hibbert, L. (2026). Shadow AI: The New Digital Frontier. *IEEE Transactions on Cyber Security*.
2. Wang, J. et al. (2025). Semantic Redaction for Generative Models. *Springer Nature*.
3. Cisco Security Report (April 2026). Trends in AI-Driven Data Exfiltration.
4. Gartner Research. (2025). Data Loss Prevention Strategies for the AI Era.

TANGIBLE DELIVERABLES

- 📄 Patent Filing: Novel semantic redaction logic.
- 📖 Journal Publication: High-impact paper on AI security.
- 🔧 Research Proposal: Secure mass production of the gateway.



THANK YOU!

Questions & Discussion

RV College of Engineering - Cyber Security 2026

IMAGE SOURCES



<https://rvce.edu.in/wp-content/uploads/2025/11/2-2.png>

Source: rvce.edu.in



https://pub.mdpi-res.com/data/data-10-00053/article_deploy/html/images/data-10-00053-g001.png?1744712139

Source: www.mdpi.com



https://miro.medium.com/1*2Zd7_u9JxONldKc6dxBzvA.jpeg

Source: medium.com



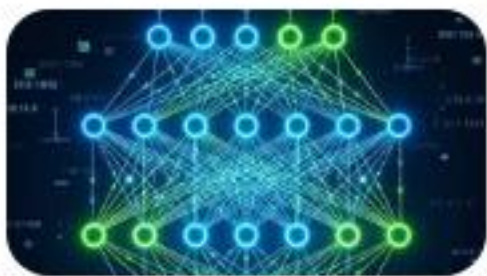
<https://updf.com/wp-content/uploads/2025/12/find-sources-for-research-paper-1.webp>

Source: updf.com



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Source: pngtree.com



<https://media.easy-peasy.ai/766d00a0-4ce3-4b53-803e-5d357ab72326/f5636542-52e0-46b8-bb4e-a0aa73a84a7a.png>

Source: easy-peasy.ai

IMAGE SOURCES



https://media.easy-peasy.ai/b0df9f99-8afc-42fa-827a-670126ab8725/b40d034b-affd-40a6-a570-82550e97b518_medium.webp

Source: easy-peasy.ai



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